

FIN ST1452–ST1541

Structured Uncertainty, Mixture E-Process,
Calibration and Refinement Robustness

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

Six research rounds — 90 programmes
Release 11.00 — 31 August 2026

This report extends the interval-certified conditional *OA* discriminator from a point decimal spectrum to a structured circulant rounding box, a common calibration polytope and an exact $12 \rightarrow 24$ coarse refinement. It also adds a valid one-sided finite-mixture e-process. None of these constructions supplies physical apparatus or selects the state ontology.

Abstract

ST1452–ST1541 build a four-level robustness ladder for the conditional FIN Operational/State discrimination protocol. First, the point decimal spectrum of Release 10.99 is replaced by a row-sum-preserving real symmetric circulant matrix uncertainty model with entry radius 5×10^{-15} . Weyl/row-sum bounds give eigenvalue radius 6×10^{-14} while preserving Fourier eigenvectors. Outward interval replay proves that every admissible spectrum has one return-gap maximum in the common bracket $[0.5945307, 0.5945309]$. The global-exclusion margin remains $3.0895722903 \times 10^{-6}$, and the four-time composite lower bound remains

$$\text{RSS} \geq 0.04669201121378897.$$

This is not a theorem for noncirculant perturbations or symbolic kernel parameters.

A numerical minimax allocation is then studied using the smaller of the two directional Bernoulli KL divergences. A finite linear programme assigns approximately 31.55% of samples to $t = 0.3$ and 68.45% to $t = 2.0$, with zero middle-time weights. Its grid objective is 1.67 times the equal- allocation value and survives a finer held-out grid. Adaptive interval lower bounds and a two-point dual upper bound certify the continuous minimax value in $[0.05, 0.05370588868619479]$. The exact optimizing weights remain open; the four-time design remains the default for the independent RSS theorem.

To repair the invalidity of nuisance likelihood refitting under optional stopping, a finite 18-component quantum nuisance mixture is frozen. Each component likelihood ratio is a martingale under the classical null; their fixed convex mixture is therefore an e-process. Crossing 100 rejects the classical null with anytime Type-I error at most one percent. The construction is one-sided and grid-scoped: it does not uniformly accept the classical model against the continuum.

A common calibration polytope with uniform preparation contamination at most 5%, false-positive/false-negative rates at most 2%, and click efficiency at least 80% preserves at least 91.2% of the ideal gap and yields

$$\text{RSS} \geq 0.03883580017500168.$$

Independent calibration maps require tighter bounds: the current proof rejects the 5% + 2% independent budget but accepts a 3% + 2% example.

Finally, the exact Kronecker-sum refinement

$$A_{24}(q) = A_{12} \otimes I_2 + I_{12} \otimes \begin{pmatrix} q & -q \\ -q & q \end{pmatrix}$$

transports every coarse return, interval certificate and likelihood exactly, independently of q . A fiber-resolving instrument adds the fine observables $(1 + e^{-2qt})/2$ and $\cos^2(qt)$, but q remains unsourced. Protocol 11.00 hashes the resulting hierarchy. Laboratory evidence remains absent.

Contents

1	Robustness ladder	3
2	Round I: ST1452–ST1466 — structured matrix uncertainty	3
3	Round II: ST1467–ST1481 — time allocation	3
4	Round III: ST1482–ST1496 — finite-mixture e-process	4
5	Round IV: ST1497–ST1511 — calibration polytopes	5
6	Round V: ST1512–ST1526 — exact refinement transport	5
7	Round VI: ST1527–ST1541 — synthesis	5
8	Recommended next programmes	6
9	Final conclusion	7

A	Six-round ledger	7
B	Reproducibility	7
C	Nonclaims	7

1 Robustness ladder

The certification hierarchy is:

1. exact frozen decimal spectrum;
2. structured row-sum-preserving circulant rounding box;
3. common preparation/detector/loss calibration polytope;
4. exact coarse $12 \rightarrow 24$ refinement.

Each level introduces explicit premises. Passing a higher level does not license arbitrary matrix uncertainty, independent instruments or a physical refinement law.

2 Round I: ST1452–ST1466 — structured matrix uncertainty

Let each independent first-row coefficient of the real symmetric circulant matrix vary within radius

$$\delta_A = 5 \times 10^{-15},$$

subject to exact row-sum zero. The perturbation remains circulant, so the Fourier eigenvectors and paired multiplicities are fixed. The eigenvalue uncertainty obeys the conservative row-sum/Weyl bound

$$|\delta \lambda_k| \leq 12\delta_A = 6 \times 10^{-14}.$$

The interval proof of Release 10.99 is repeated with these eigenvalue boxes. Outside $[0.59, 0.60]$, the largest Taylor-cover upper bound is 0.41123791023528017, while the uncertain candidate lower bound is 0.41124099980757045. The margin is

$$3.089572290282394 \times 10^{-6} > 0.$$

On $[0.59, 0.60]$,

$$F'' \in [-1.686445990051945, -1.3685959622934722],$$

so every admissible return-gap curve is strictly concave. Uniform derivative signs at 0.5945307 and 0.5945309 prove that each curve has one stationary root in that common bracket. Consequently every admissible spectrum has its global maximum inside $[0.59, 0.60]$, and its root lies in $[0.5945307, 0.5945309]$. A common value enclosure has width about 2.43×10^{-13} .

The complete 1000-cell nuisance cover also survives:

$$\text{RSS} \geq 0.04669201121378897, \text{quad} \max_i |r_i| \geq 0.10804167160613187.$$

Theorem 1 (structured uncertainty persistence). *The unique time-optimum location and positive four-time composite separation remain valid throughout the declared symmetric circulant rounding box.*

The theorem does not cover noncirculant errors, eigenvector rotation or a box in $(\omega, \phi, \beta, \eta)$.

3 Round II: ST1467–ST1481 — time allocation

For times $T = (0.3, 0.6, 1.2, 2.0)$ and nuisance parameter $\theta = (\alpha, \gamma)$, define the per-time symmetric design score

$$d_i(\theta) = \min\{D(C_i \| Q_{i,\theta}), D(Q_{i,\theta} \| C_i)\}.$$

The finite design problem is

$$\max_{w_i \geq 0, \sum_i w_i = 1} \min_{\theta \in \Theta_{\text{grid}}} \sum_i w_i d_i(\theta).$$

On a 41×201 training grid, linear programming returns

$$w = (0.3155394972, 0, 0, 0.6844605028),$$

with objective 0.0537067608. Equal allocation gives 0.0322200827, so the grid improvement factor is about 1.667. A finer independent 81×401 grid has minimum 0.0537030928 near $(\alpha, \gamma) = (1.1, 16.25)$.

For 100 integer shots, the rounded candidate is $(32, 0, 0, 68)$. A separate outward interval cover proves a positive weighted squared-residual lower bound 0.0046819465 for these two endpoint times. Thus exact matching is excluded on the declared nuisance box.

An adaptive outward cover then subdivides only cells whose analytic rectangular KL lower bound is below 0.05. After 380 splits, 1380 accepted boxes cover the entire continuous nuisance rectangle and prove that the candidate allocation has value at least 0.05. Conversely, a fixed dual mixture supported at $(1.1, 16)$ and $(1.1, 16.25)$ bounds the value of *every* allocation by 0.05370588868619479. Therefore the continuous minimax value for the declared metric satisfies

$$0.05 \leq V_* \leq 0.05370588868619479.$$

The exact maximizing weights remain open. The grid candidate is a certified near-minimax design, not an exact continuous optimizer. Its worst case lies on the clock boundary and the solution lies on a two-time face, making it sensitive to model-library expansion. The four-time design remains the certified conservative default for the independent RSS theorem.

4 Round III: ST1482–ST1496 — finite-mixture e-process

The alternative grid contains

$$\alpha \in \{0.9, 1.0, 1.1\}, qquad \gamma \in \{0, 5, 10, 20, 50, 100\},$$

for 18 components with uniform weights $\pi_j = 1/18$. For component j let

$$L_{j,n} = \prod_{i=1}^n \frac{q_j(X_i|t_i)}{p_C(X_i|t_i)}.$$

Under the classical null every $L_{j,n}$ is a nonnegative mean-one martingale, and hence

$$E_n = \sum_{j=1}^{18} \pi_j L_{j,n}$$

is one.

Theorem 2 (one-sided mixture test). *Under the frozen classical null,*

$$\Pr_C \left(\sup_n E_n \geq 100 \right) \leq 0.01.$$

The conclusion holds for optional stopping under the predeclared schedule.

If component j is true, the mixture obeys $E_n \geq \pi_j L_{j,n}$ and initially pays at most $\log 18$ in log-evidence. A seeded grid-alternative fixture crossed the boundary after 32 events; a 500-event classical fixture did not. These are software tests.

The e-process only supports rejection of C in favour of the frozen mixture. It does not provide uniformly valid acceptance of C against every component, and off-grid power is unproved. A continuous prior mixture is a separate construction.

5 Round IV: ST1497–ST1511 — calibration polytopes

The common binary calibration map is

$$T_{\epsilon,a,b}(p) = a + (1 - a - b) \left[(1 - \epsilon)p + \frac{\epsilon}{12} \right],$$

with

$$0 \leq \epsilon \leq 0.05, 0 \leq a, b \leq 0.02.$$

When the same map applies to both hypotheses, every return gap is scaled by

$$g = (1 - a - b)(1 - \epsilon) \geq 0.96 \cdot 0.95 = 0.912.$$

Therefore the base gap remains at least 0.3750310, and the four-time certificate becomes

$$\text{RSS} \geq g^2(0.04669201121378897) = 0.03883580017500168.$$

At least one residual remains above 0.0985340045.

If calibration maps may differ by hypothesis, the proof uses the per-model bound

$$e_{\text{model}} \leq \frac{11}{12} \epsilon_{\text{prep}} + \delta_{\text{detector}}.$$

To preserve the certified one-time residual, this must be below $0.1080416716/2 = 0.0540208358$. The independent budget $\epsilon = 0.05$, $\delta = 0.02$ gives 0.0658333 and is not certified. The example $0.03 + 0.02$ gives 0.0475 and passes. Failure of the sufficient bound does not prove actual nonidentifiability.

Known uniform click efficiency $\eta \in [0.8, 1]$ preserves click-conditioned vertex distributions but requires attempts and no-clicks. The mean attempt inflation at $\eta = 0.8$ is 1.25; this is not a tail guarantee. Vertex-dependent efficiencies require a stochastic/substochastic detector matrix polytope and remain open.

6 Round V: ST1512–ST1526 — exact refinement transport

Let

$$B_q = \begin{pmatrix} q & -q \\ -q & q \end{pmatrix}, \text{quad} A_{24} = A_{12} \otimes I_2 + I_{12} \otimes B_q.$$

The commuting Kronecker sum gives

$$e^{-tA_{24}} = e^{-tA_{12}} \otimes e^{-tB_q}, \text{quad} e^{-itA_{24}} = e^{-itA_{12}} \otimes e^{-itB_q}.$$

If the instrument sums over the two fiber outcomes at a base node, both classical and quantum node-return probabilities equal their 12-state values for every $q \geq 0$ and either initial fiber basis state. Consequently the time-optimum certificate, four-time RSS certificate and seven-distance-class likelihood transport exactly. The coarse protocol is q -blind.

A fiber-resolving measurement supplies additional returns for initial fiber state 0:

$$H_q(t) = \frac{1 + e^{-2qt}}{2}, \quad R_q(t) = \cos^2(qt).$$

This creates a fine dual-dynamics discriminator. It requires a 24-outcome or fiber-resolving calibrated detector and does not derive q or its physical units.

7 Round VI: ST1527–ST1541 — synthesis

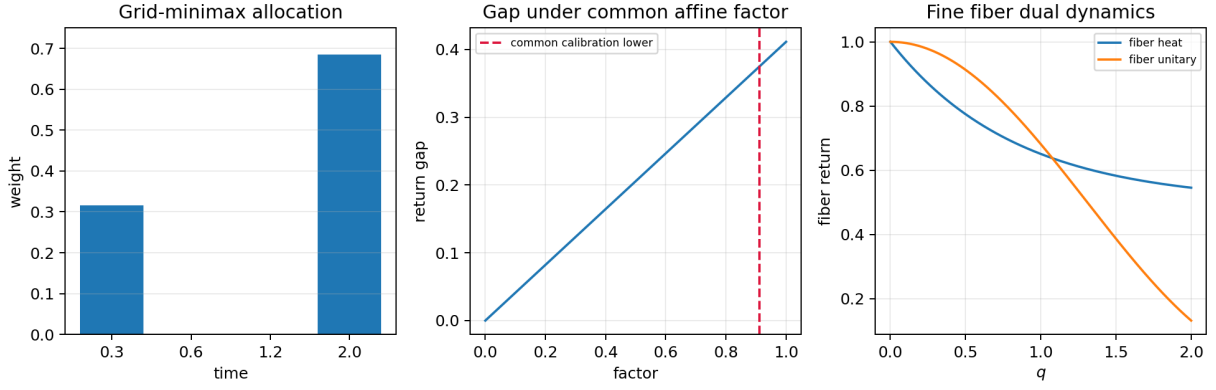


Figure 1: Left: numerical grid-minimax time weights. Centre: return-gap scaling under a common affine calibration factor. Right: the optional fine-fiber heat and unitary returns at $t = 0.6$.

Layer/result	Status	Boundary
structured circulant uncertainty	[Interval-proven]	radius 5×10^{-15} , Fourier structure fixed
KL time allocation	[Certified value bracket]	exact optimizing weights open
two-time nonidentity	[Interval-proven]	weighted squared metric, declared box
finite-mixture e-process	[Proven one-sided]	18 frozen components
common calibration polytope	[Proven]	common affine map
independent calibration	[Conditional bounds]	3% + 2% example passes
coarse 12 $\rightarrow$ 24 transport	[Exact]	declared Kronecker refinement
fine fiber discriminator	[Constructed]	q and instrument unsourced
laboratory evidence	[Absent]	no apparatus/custody/events

Protocol `FIN_ST1527_ST1541_Protocol_11_00.json` hashes the parent protocol, mixture grid and calibration polytope, and records the structured uncertainty, interval bounds, numerical allocation and refinement rule.

These results make one conditional *OA* discriminator robust across four explicit layers. They do not derive a state category, strict clock, physical detector or the value q .

The protocol still tests internal mixing versus coherent dynamics. It does not test literal nonexistence of the whole nadsoliton.

8 Recommended next programmes

1. derive interval eigenvector bounds for noncirculant matrix boxes;
2. propagate symbolic $(\omega, \phi, \beta, \eta)$ uncertainty;
3. narrow the continuous KL value bracket and certify optimizing weights;
4. construct a continuous-prior mixture e-process;
5. optimize an independent calibration polytope;
6. certify a vertex-dependent detector matrix box;
7. optimize integer shot allocation;
8. test fine-fiber identifiability of q ;

9. impose multi-level refinement consistency;
10. derive one platform-specific forward model without claiming realization;
11. freeze external calibration and custody forms;
12. obtain independent review before raw data collection.

Stop if a noncirculant error is treated as circulant, a grid optimum is called a continuum theorem, mixture rejection is called symmetric decision, a common calibration proof is applied to independent devices, coarse transport is called a derivation of q , or a synthetic fixture is called evidence.

9 Final conclusion

The conditional FIN dual-dynamics discriminator is now stable against one declared structured matrix box, one common calibration polytope and one exact refinement family. A finite nuisance mixture also permits valid anytime rejection of the classical simple null. The principal remaining mathematical gap is robustness to noncirculant/eigenvector uncertainty and continuous nuisance families; the principal physical gap remains an independently calibrated apparatus and raw event record.

A Six-round ledger

Range	Object	Verdict
ST1452–ST1466	structured matrix uncertainty	interval certificates survive
ST1467–ST1481	time allocation	continuous value bracket; exact weights open
ST1482–ST1496	mixture e-process	valid one-sided finite-grid rejection
ST1497–ST1511	calibration polytope	common robustness; independent bound split
ST1512–ST1526	refinement transport	exact coarse invariance, fine test
ST1527–ST1541	synthesis	four-level conditional robustness ladder

B Reproducibility

The authoritative result sets run from `FIN_ST1452_ST1466_Results.json` through `FIN_ST1527_ST1541_Results.json`. Each contains fifteen programmes and packet hashes. The combined test `test_fin_st1452_st1541.py` verifies all ninety packets, uncertainty margins, allocation status, mixture fail-closed behaviour, calibration gates, refinement identities, protocol and figure.

C Nonclaims

No arbitrary-matrix or symbolic-kernel interval theorem, continuous minimax solution, continuous composite e-process, platform calibration, state ontology, apparatus, raw event, independent custody, legacy bridge, selector, Standard Model, gravity, L_{total} or Theory of Everything is established. All future FIN PDFs remain English.

References

- [1] R. E. Moore, R. B. Kearfott and M. J. Cloud, *Introduction to Interval Analysis*, SIAM.
- [2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley.
- [3] J. Ville, *Etude critique de la notion de collectif*, Gauthier-Villars.